SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.*(BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

*I **P Dinesh Reddy / 192525399** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	**Smart Healthcare Diagnosis**	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	**Student Academic Advising**	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.

3	**Intelligent Loan Approval**	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.
		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	**Vehicle Fault Diagnosis**	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.

5	**Cybersecurity Threat Detection**	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

Vehicle Fault Diagnosis

Problem Analysis & Knowledge Identification

Entity: Car101

Constant: Car101

Facts

- Car101 does not start.
- Car101 has dim headlights.

Rules

1. DoesNotStart ∧ DimHeadlights → BatteryProblem
2. BatteryProblem → BatteryInspection
3. BatteryInspection → Service

Propositional Logic

Let:

- D = Car101 does not start
- H = Car101 has dim headlights
- B = Car101 has a battery problem

- I = Car101 requires battery inspection
- S = Car101 is marked for service

Facts

$D \land H$

Rules

$(D \land H) \rightarrow B$

$B \rightarrow I$

$I \rightarrow S$

Inference

$D \land H \rightarrow B \rightarrow I \rightarrow S$

Therefore, **Car101 has a battery problem and should be marked for service.**

FOL Representation

Predicates

- DoesNotStart(x)
- DimHeadlights(x)
- BatteryProblem(x)
- BatteryInspection(x)
- Service(x)

Facts

DoesNotStart(Car101)

DimHeadlights(Car101)

Rules

∀x [(DoesNotStart(x) ∧ DimHeadlights(x)) → BatteryProblem(x)]

∀x [BatteryProblem(x) → BatteryInspection(x)]

∀x [BatteryInspection(x) → Service(x)]

Predicates, Variables, Constants & Quantifiers

Component	Representation
Constant	Car101
Variable	x

Component Representation

Predicates	DoesNotStart(x), DimHeadlights(x), BatteryProblem(x), BatteryInspection(x), Service(x)
Quantifier	∀x – for every vehicle

Logical Connectives

- **AND (∧):** DoesNotStart(x) ∧ DimHeadlights(x)
- **Implication (→):** (DoesNotStart(x) ∧ DimHeadlights(x)) → BatteryProblem(x)
- **Implication (→):** BatteryProblem(x) → BatteryInspection(x)
- **Implication (→):** BatteryInspection(x) → Service(x)

Unification & Substitution

Consider:

DoesNotStart(x)

and

DoesNotStart(Car101)

The substitution is:

θ = {x / Car101}

Therefore:

DoesNotStart(x)θ = DoesNotStart(Car101)

Similarly:

DimHeadlights(x) and DimHeadlights(Car101)

MGU = {x / Car101}

Resolution / Proof Construction

Convert the first rule into clause form:

¬DoesNotStart(x) ∨ ¬DimHeadlights(x) ∨ BatteryProblem(x)

Given:

1. DoesNotStart(Car101)
2. DimHeadlights(Car101)

Using **x = Car101**:

BatteryProblem(Car101)

Next rule:

$\neg$BatteryProblem(x) $\lor$ BatteryInspection(x)

Therefore:

BatteryInspection(Car101)

Next rule:

$\neg$BatteryInspection(x) $\lor$ Service(x)

Therefore:

Service(Car101)

Complete Inference Sequence

DoesNotStart(Car101)

+

DimHeadlights(Car101)

↓

BatteryProblem(Car101)

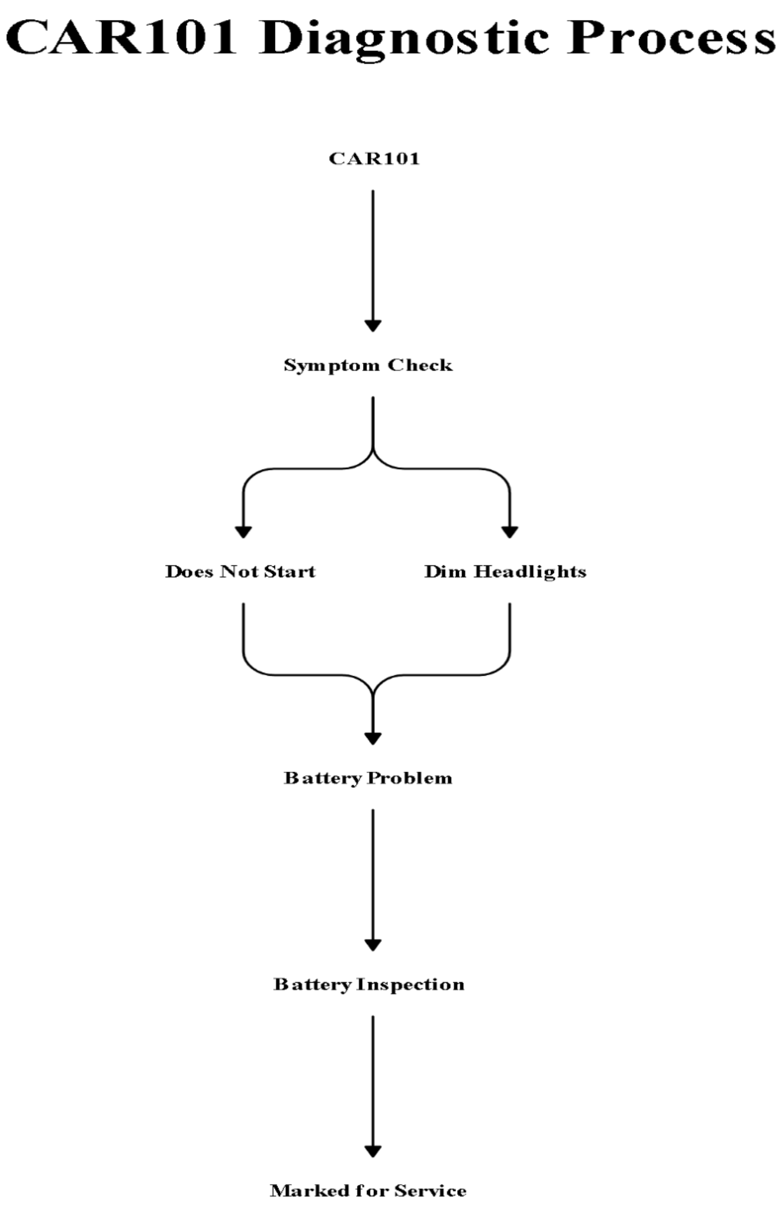
↓

BatteryInspection(Car101)

↓

Service(Car101)

Mapping Diagram



Conclusion

Car101 has a battery problem. Therefore, it requires battery inspection and should be marked for service.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
Inference Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.

Resolution Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.